

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021****Subject Code:3161007****Date:02/12/2021****Subject Name:Computer Networks****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) A bit string 011110111110111110, needs to be transmitted at the data link layer. What is the string actually transmitted after bit stuffing?	03
	(b) Which of the OSI layers handle each of the following:	04
	(1) Dividing the transmitted bit stream into frames	
	(2) Determining which route through the subnet route	
	(3) Dividing the application data into segments	
	(4) Give some examples of protocols that work at the Application layer?	
	(c) What is the difference between error detection and correction? Explain any one error correction technique with suitable example	07
Q.2	(a) What is the difference between port address, logical address and physical address?	03
	(b) What are the responsibilities of the network layer in the Internet model?	04
	(c) Explain IPv4 datagram format and importance of each field	07
	OR	
	(c) Explain CSMA/CD protocol in detail.	07
Q.3	(a) What is count-to-Infinity problem? Explain it with example	03
	(b) Write a short note on Network classes. What is the use of subnetting in assigning IP address?	04
	(c) What is the difference between hub and a switch? In which layer of OSI model do each of these operate? Discuss the use of router and bridge in computer networks.	07
	OR	
Q.3	(a) Distinguish between Fast Ethernet and Gigabit Ethernet standards	03
	(b) Answer the following in brief:	04
	(1) How ARP resolves IP address into MAC address?	
	(2) What do you mean by piggybacking?	
	(c) What is ALOHA? Explain variants of ALOHA protocol.	07
Q.4	(a) Differentiate between static and dynamic routing.	03
	(b) Explain the concept of flooding.	04
	(c) With a neat diagram explain distance vector routing protocol	07
	OR	
Q.4	(a) Explain leaky bucket and token bucket algorithms?	03
	(b) How does NAT works? Explain	04
	(c) Explain the three different phases in a TCP transmission with the help of diagrams.	07

- Q.5 (a)** What are the issues of stop and wait protocol at data link layer? How selective repeat protocol resolves issues of stop and wait protocol? **03**
- (b)** Write a short note on RSA cryptosystem **04**
- (c)** What is the role of Domain Name Server (DNS) in Internet? Explain the hierarchy of various domain names. **07**

OR

- Q.5 (a)** Write a short note on: ICMP **03**
- (b)** Distinguish between symmetric and asymmetric encryption scheme. **04**
- (c)** Differentiate the following: **07**
- (i) Connection Oriented Services and Connection Less Services (ii) OSPF and BGP
